



Rensselaer

Course Information

Course Title: Internetworking of Things
Course Number: ECSE-4660/ECSE-6660
CRN: 32409/ 32410
Credit Hours: 3
Semester / Year: Spring 2026
Meeting Days: MR: 12:30-1:50 PM
Lecture Room Location: JONSSON 4104
Laboratory Location: JEC 5312
Laboratory Time: 7 AM – 10 PM (access provided to registered students)

Websites/Platforms:

- Piazza: for accessing lecture material and discussions.
- Gradescope: For submitting assignments
- LMS: for Quizzes and Homeworks
- Webex: Project groups

Prerequisites: Basic understanding of computer organization, operating systems & networks. Ability to write computer programs in high-level languages (such as Python or C/C++).

Instructors

Professor Ish Jain
Office Hours: Mondays and Thursdays after class or by appointment.
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Teaching Assistants

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Office/Lab Hours: Mon-Fri 4-5 PM

Course Description

The course provides an in-depth study of the technologies and protocols involved in building the Internet-of-Things (IoT), with specific focus on networking at the edge of the Internet. This includes understanding of wireless communication and link layer technologies, multi-access and scheduling mechanisms, mobility models, routing in disconnected networks, energy-efficient edge networking, loss tolerant transport protocols, and their applications to emerging areas such as vehicular networks, RFID systems and smart buildings. The course also discussed IoT Security and data/content distribution, aggregation, and compression. This course has a strong emphasis on **hands-on experience** utilizing Raspberry Pi's, Arduino's, and NI software radio boards, and a significant part of the course assessment will be based on a **final project** focused on building a

wireless based application such as indoor localization for IoT devices. Students are encouraged to suggest their own individual projects.

Topics to be covered include

1. Review of networking architectures and standards
2. Review of wireless PHY layer technologies
3. Wireless access technologies for IoT: WiFi, LTE, NB-IoT
4. Low-power IoT Wireless Technologies: ZigBee, Z-Wave, BLE, LoRA
5. Low-power Backscatter Technologies: RFID
6. Routing and network layer protocols for IoT: RPL and 6LoWPAN
7. Low-overhead IoT transport: CoAP and MQTT
8. Network management for IoT: LWM2M and NETCONF
9. IoT mobility models and mobility handling protocols
10. IoT Security
11. Energy efficient IoT networking
12. Sensor data aggregation and analytics
13. Applications to vehicular networking, environmental monitoring, healthcare, smart buildings etc.

Course Texts

No required textbooks. This is an advanced-level course, and much of the material covered in this course may not be available in textbook form. Reference material will be provided for each lecture.

Supplemental textbooks list (not required):

1. Olivier Hersent, David Boswarthick and Omar Elloumi, The Internet of Things: Key Applications and Protocols, 2nd ed., Wiley, 2012.
2. Holger Karl and Andreas Willig, Protocols and Architectures for Wireless Sensor Networks, 1st ed., Wiley, 2007.
3. Vijay Garg, Wireless Communications and Networking, Morgan Kauffman, 2007.
4. Savo Glisic, Advanced Wireless Communications and Internet: Future Evolving Technologies, 3rd ed., 2011.
5. Jack L. Burbank, Julia Andrusenko, Jared S. Everett and William T.M. Kasch, Wireless Networking: Understanding Internetworking Challenges, Wiley, 2013.

Student Learning Outcomes

The table below presents the Student Learning Outcomes (SLOs) for both ECSE-4660 and ECSE-6660 students. The left column is for ECSE 4660 students, while the right column for ECSE-6660 students expands each with greater depth, advanced applications, and evaluation criteria.

ECSE-4660	ECSE-6660
1. Understand, design, and evaluate wireless PHY layer technologies for IoT such as Modulation, LoRa, RFID, etc.	1. Analyze and compare advanced wireless PHY layer technologies (e.g., LoRa, NB-IoT, UWB), including performance trade-offs, spectral efficiency, and deployment considerations in diverse IoT environments.

ECSE-4660	ECSE-6660
2. Understand key wireless access technologies such as Aloha and CSMA used in the Internet-of-Things.	2. Critically evaluate and simulate advanced MAC protocols (e.g., TDMA, FDMA, SDMA) for IoT, considering scalability, latency, and energy efficiency in dense and heterogeneous networks.
3. Understand key routing algorithms and protocols used in the Internet-of-Things.	3. Design and assess routing strategies (e.g., RPL, geographic routing, opportunistic routing) under constraints such as mobility, energy, and intermittent connectivity in large-scale IoT deployments.
4. Understand low-overhead transport protocols that are suited for IoT applications.	4. Evaluate and optimize lightweight transport protocols (e.g., CoAP, MQTT, QUIC) for reliability, congestion control, and interoperability in constrained IoT networks.
5. Understand mobility models and mobility handling protocols for the Internet-of-Things.	5. Model and simulate mobility patterns in IoT (e.g., vehicular, drone-based, or mobile health networks), and assess the impact on routing, latency, and data consistency.
6. Understand security mechanisms used for IoT communication over wireless.	6. Analyze and implement advanced security frameworks for IoT, including lightweight cryptographic protocols, secure bootstrapping, and intrusion detection in resource-constrained environments.
7. Understand data compression and aggregation methods in sensor IoT networks.	7. Design and evaluate adaptive data compression, in-network processing, and edge analytics techniques for scalable and energy-efficient IoT data management.
8. Understand integrated IoT solutions critical in emerging domains such as vehicular networks, RFID systems, and smart buildings.	8. Architect and critically assess end-to-end IoT systems in complex domains (e.g., smart cities, industrial IoT, cyber-physical systems), focusing on interoperability, scalability, and real-time performance.

Note that in addition, students at 6000 level will implement practical security considerations in IoT networks through advanced labs and individual projects.

Course Assessment Measures

The assessment measures include regular homeworks, assignments, and a final project. The weightage of each assessment measure (as a percentage of the total course grade) is as follows:

Assessment Component	ECSE-4660	ECSE-6660
Class Participation	5%	5%
Per-class Quizzes	5%	5%
Homework Assignments	20%	20%
Laboratory Assignments	30%	30%
Final Project (Implementation and Demonstration)	40%	40%

- The grade distribution above is the same for both 4660 and 6660 students. **For students signed up for ECSE-6660, some advanced lab/homework questions and a more advanced project are required.**
- **Projects are done in a group of two students. Students at 4000 and 6000 level are discouraged to form a group together, unless approved by the instructor, and in this case, the student at 6000 level is expected to work on the advanced features on its own.**

Course Topics and Schedule (Tentative)

Since this is an advanced topics course, the following schedule and the list of topics are tentative, and subject to change depending upon the progress of the class.

Lecture #	Topic
1,2	Intro to network architectures & standards
3,4,5,6	PHY layer technology review: channel, LoRA, backscatter, OFDM foundation
7,8,9,10	Wireless access technologies for IoT: WiFi, BLE, LTE, Zigbee
11,12,13, 14,15	Routing and network layer protocols for IoT: DSR, AODV, IP addressing, DHCP, ARP, IPv6, NAT, Routing, ICMP, RPL. 6LowPAN
16,17	Low-overhead IoT transport: UDP, TCP, Socket Programming
18,19	Applications for IoT: HTTP, MQTT, CoAP
20,21,22, 23	Secure communication of IoT devices over wireless; Secure protocols, firewall
24,25	Emerging applications
26,27,28	Final project presentations

Other Course Policies

Final Project:

The project provides an opportunity for the student to immerse themselves in the design and application of IoT in their favorite area of interest. To favor creativity, critical thinking, brainstorming, and team organization, each project will typically involve **up to 2 students** (*doing the project individually is fine, but if you want to do the project in a group of 3 or more students, you need to provide a strong justification why so many members are needed*). The final project report will be submitted at the end of the semester, together with a presentation, and a recorded demo. Any

software code will need to be uploaded with appropriate documentation in order for the instructor to check by re-generating the results. More details about the project will be shared throughout the class.

Important: Students cannot use a previously developed product, whether by them in another class, or through a third party, towards getting credit for the final project. Any open-source or third-party software needs to be explicitly acknowledged and will not count towards credit. Failing to acknowledge such open/third-party software will be considered plagiarism and will be handled according to our academic dishonesty policy (see below).

Quiz/Homework/Lab Assignments:

- Per-class Quizzes will be **assigned on LMS** before every lecture (Lecture 2 onwards) and will be constitute of simple questions that will require you to pay attention to the material discussed in class. They will be due the end of day (11:59pm on the same day).
- **Homework/Lab assignments will be on Piazza.** Students need to submit their solutions on Gradescope by the deadline. These assignments will be posted roughly on a weekly basis, with homeworks and lab assignments due in alternate weeks.
- **Late assignment submissions** will not be accepted, unless approved by the instructor. There is a 20% penalty until late submission deadline on Gradescope (typically 2 days later). First two penalties will be waived. LMS grading is automatic and doesn't allow late submissions.
- All assignments are to be **completed on your own**. You are allowed to consult with other students in the current class or the instructor (e.g. through Piazza) regarding the conceptualization of the problem and possible methods of solution, but you may not share details or your final answer with other students. You may not copy solutions from anyone or any source. Failure to follow the rules will be considered violation of academic integrity.
- You may not knowingly provide your work to be copied. Doing so will also be considered violation of academic integrity as well.

Attendance/Participation Policy:

We believe that effective learning is best supported by a climate of inquiry, in which students are actively engaged in the learning process. While synchronous lecture attendance is not mandatory, it is important to performing well in this course for all students to attend/watch the lecture and promote a climate of inquiry and active involvement in the learning process.

Submission Policy and Absences:

A job interview is not an approved absence unless approved beforehand, with appropriate note from the Dean of Students. An approved absence is one that has been approved following Rensselaer's rules and procedures, and should be documented, typically via a letter from the dean's office, and emailed beforehand to the instructor.

- All *homework* and *lab* assignments must be uploaded *before the deadline date on Gradescope*. Late submissions are automatically marked and not accepted. Without a valid excuse, a grade of zero is given for a late/missing homework.
- The *final project* report and associated deliverables must be submitted by the final class. In case of valid excuse, a grade of incomplete will be given for the course, otherwise a grade of zero will be assigned.

Academic Integrity

Student-teacher relationships are built on trust. For example, students must trust that teachers have made appropriate decisions about the structure and content of the courses they teach, and teachers must trust that the assignments that students turn in are their own. Acts that

violate this trust undermine the educational process.

The Rensselaer Handbook of Student Rights and Responsibilities and the Graduate Student Supplement (For 6000 level and above courses) define various forms of Academic Dishonesty and you should make yourself familiar with these. In this class, all assignments that are turned in for a grade must represent the student's own work. In cases where help was received, or teamwork was allowed, a notation on the assignment should indicate your collaboration. Submission of any assignment that is in violation of this policy will result in (1) an academic (grade) penalty and (2) reporting to Associate Dean of Academic Affairs and either the Dean of Students (for Undergraduates) or the Dean of Graduate Education (for Graduate students). In this course, the academic penalty for a first offense is zero grade for the relevant portion of the grade. A second offense will result in failure of the course.

If you have any questions concerning this policy before submitting an assignment, please ask for clarification.

Academic Accommodations

Rensselaer Polytechnic Institute strives to make all learning experiences as accessible as possible. If you anticipate or experience academic barriers based on a disability, please let me know immediately so that we can discuss your options. To establish reasonable accommodations, please register with The Office of Disability Services for Students (mailto:dss@rpi.edu; 518-276-8197; 4226 Academy Hall). After registration, make arrangements with me as soon as possible to discuss your accommodations so that they may be implemented in a timely fashion.”